COMPETENCY-BASED CURRICULUM FRAMEWORK • CORE COMPETENCIES

MODULE 9: OPERATING A SPREADSHEET APPLICATION

CBC-BCL-09
Rev. 00-03/08/17

UNIT OF COMPETENCY	NOMINAL DURATION	PREREQUISITES
Operate a Spreadsheet Application	18 Hours (Modular / Hands-On)	Operating a Personal Computer

Module Descriptor: This module defines the competency required to operate spreadsheet applications (such as Microsoft Excel / LibreOffice Calc) and perform computational operations. It covers creating and managing workbooks, data type entry, utilizing mathematical formulas and built-in functions, formatting cells and sheet layouts, generating visual charts, and executing customized print operations.

SUMMARY OF CORE LEARNING OUTCOMES:

LO 1	LO 2	LO 3	LO 4
Create & Save Worksheets	Customize Settings & Format	Incorporate Objects & Charts	Page Setup & Print Worksheets

LO 1. CREATE AND SAVE WORKSHEETS

CORE LEARNING OUTCOME 1

ASSESSMENT CRITERIA	TECHNICAL CONTENTS	CONDITIONS & RESOURCES	METHODOLOGIES	ASSESSMENT
1. Data Entry & Types: Text, numeric values, dates, time, and currency entered, recognized, and edited within worksheet cells according to task requirements.	• Spreadsheet anatomy (Row, Column, Cell) • Data types: General, Number, Currency, Date • AutoFill handle & sequential series • In-cell editing & Formula Bar controls	• PC workstation with MS Excel • Sample raw business data sets • Numeric keypad peripherals • Trainee exercise manuals	• Lecture-demonstration • Hands-on application • Self-paced tasks	• Direct observation • Practical demonstration • Computer-based test
2. Formula Construction: Basic mathematical operators (+, −, *, /) and standard functions (SUM, AVERAGE, MIN, MAX, COUNT) authored and debugged accurately.	• Formula syntax starting with '=' • Relative vs. Absolute cell references (A1) • Common error values (#DIV/0!, #VALUE!, #REF!) • Order of operations (PEMDAS)	• Inventory and payroll practice sheets • Formula reference cheat-sheets • Calculation problem cards	• Guided problem-solving • Practical laboratory drill • Interactive review	• Skills test • Formula correctness audit • Direct observation
3. Workbook Storage & Retrieval: Workbooks saved with specified naming standards in designated directories, flash drives, or cloud storage; existing sheets opened.	• Workbook vs. Worksheet concepts • File formats (.xlsx, .xls, .csv, .pdf) • Workbook sheet tab controls (Add, Rename, Color) • AutoRecover and backup options	• Workstation file system access • USB Flash drive storage • Organizational filing rubric	• Demonstration • Hands-on task simulation	• File storage inspection • Oral questioning

LO 2. CUSTOMIZE BASIC SETTINGS AND FORMAT WORKSHEETS

CORE LEARNING OUTCOME 2

ASSESSMENT CRITERIA	TECHNICAL CONTENTS	CONDITIONS & RESOURCES	METHODOLOGIES	ASSESSMENT
1. Row & Column Layout: Column widths and row heights modified; rows and columns inserted, deleted, hidden, unhidden, and auto-fitted to text.	• Adjusting column widths & row heights • AutoFit column width controls • Inserting & deleting multiple rows/columns • Hiding/unhiding confidential data rows	• Workstations with spreadsheet apps • Layout practice templates • Standard tabular guidelines	• Demonstration • Hands-on lab exercises • Discussion	• Direct observation • Practical demonstration • Skill rubric check
2. Cell Typography & Alignment: Font style, size, color, cell text wrapping, Merge & Center, vertical alignment, and cell orientations adjusted.	• Font settings for financial headers • Merge & Center and Wrap Text tools • Text orientation & angled column headers • Vertical and horizontal cell alignments	• Model expense report sheets • Typography reference guides • Unformatted ledger data	• Guided walkthrough • Individual laboratory work	• Practical performance test • Output appearance audit
3. Numeric & Accounting Formatting: Currency symbols (₱, $), decimal places, thousands separators (,), percentages, and dates formatted per accounting rules.	• Number format dropdown & shortcuts • Accounting vs. Currency style alignment • Decimal place increase/decrease controls • Negative number display styles (parentheses/red)	• Accounting exercise worksheets • Currency style standards • Financial task prompt cards	• Practical drill • Self-paced modules	• Computer-based exam • Accuracy verification
4. Borders, Shading & View Controls: Gridlines, cell borders, accent fill colors, Freeze Panes, and multiple window split views configured cleanly.	• Cell border weights, double accounting lines • Background fill patterns and themes • Freeze Panes (Top Row, First Column) • Viewing multiple sheets side-by-side	• Multi-page ledger data sets • Screen layout standards • Practice workbooks	• Lecture-demonstration • Hands-on exercise	• Direct observation • Skill evaluation

Freeze Panes Best Practice: In large operational spreadsheets spanning hundreds of rows and columns, trainees must utilize Freeze Panes under the View tab. Freezing top header rows preserves immediate column context as operators scroll downward through inventory and payroll logs.

✓ Merge & Center ✓ Wrap Text ✓ Accounting Formats ✓ Freeze Panes ✓ AutoFit Columns

LO 3. INCORPORATE OBJECTS AND CHARTS IN WORKSHEET

CORE LEARNING OUTCOME 3

ASSESSMENT CRITERIA	TECHNICAL CONTENTS	CONDITIONS & RESOURCES	METHODOLOGIES	ASSESSMENT
1. Chart Generation: Data ranges highlighted accurately; appropriate 2D/3D Column, Bar, Line, or Pie charts created matching reporting needs.	• Chart types & data comparison principles • Selecting continuous vs. non-adjacent data (Ctrl key) • Column, Bar, Line, Pie, and Area chart selection • Recommended Charts utility	• Workstations with spreadsheet apps • Tabular quarterly sales data sets • Chart selection guide	• Lecture-demonstration • Hands-on chart creation • Interactive review	• Demonstration of skills • Practical performance test • Direct observation
2. Chart Customization & Formatting: Chart titles, axis titles, data labels, legends, gridlines, and color palettes applied and formatted cleanly.	• Chart Elements panel (Titles, Labels, Legend) • Chart Design & Format ribbons • Switching Row/Column data orientations • Adjusting chart area and plot area fills	• Model corporate chart designs • Style requirement rubrics • Trainee practice workbooks	• Step-by-step walkthrough • Guided individual drill	• Output inspection • Visual appeal & accuracy check
3. Chart Relocation & Resizing: Charts repositioned cleanly on existing worksheets or moved to dedicated Chart Sheets; aspect ratio preserved.	• Moving charts: Object in sheet vs. Chart Sheet • Resizing handles and locking proportions • Alignment with underlying gridlines (Alt key)	• Spreadsheet software • Layout specification cards • Sample dashboard templates	• Demonstration • Hands-on exercises	• Direct observation • Skill rubric review
4. Object & Shape Incorporation: Corporate logos, illustrative shapes, text callouts, and organizational diagrams inserted and aligned with tabular data.	• Inserting pictures and corporate logos • Drawing shapes, arrows, and KPI callout banners • Object grouping and layering (Bring Forward/Send Backward)	• Image asset practice library • KPI dashboard exercises • Instructional handouts	• Practical lab tasks • Self-paced tutorial	• Portfolio review • Computer examination

Chart Honesty & Data Ethics: Trainees must be taught to maintain zero-baseline honesty on bar and column charts. Truncated axes that exaggerate trivial fluctuations distort business reality and violate professional presentation standards.

✓ Column & Bar Charts ✓ Trend Lines ✓ Data Labels & Legends ✓ Dedicated Chart Sheets ✓ Non-Adjacent Data Selection

LO 4. PRINT WORKSHEETS

CORE LEARNING OUTCOME 4

ASSESSMENT CRITERIA	TECHNICAL CONTENTS	CONDITIONS & RESOURCES	METHODOLOGIES	ASSESSMENT
1. Print Area Configuration: Specific cell ranges designated as Print Area; print areas cleared, expanded, or inspected via Page Break Preview.	• Set Print Area vs. Clear Print Area commands • Page Break Preview mode and boundary dragging • Page orientation (Landscape for wide sheets)	• Wide multi-column raw data sets • Workstation with printer connectivity • Print boundary task sheets	• Lecture-demonstration • Hands-on laboratory drill	• Practical demonstration • Direct observation
2. Page Setup & Sheet Options: Scaling (Fit to 1 page wide), margins, repeating top header rows ($1:$1) across pages, and printing gridlines enabled.	• Page Setup: Fit to 1 Page Wide by X Pages Tall • Rows to Repeat at Top (Print Titles) • Printing gridlines, row and column headings • Custom Headers & Footers (Page numbers, Sheet name)	• Multi-page accounting ledgers • Printing policy rubrics • Standard bond paper	• Guided simulation • Individual practice drill	• Print preview audit • Practical test
3. Print Preview & Output Submission: Worksheets previewed to eliminate orphaned rows; printed cleanly to physical printer or exported to digital PDF.	• Print Preview inspection protocol • Printer selection, duplex printing, collation • Exporting pristine worksheets to PDF format • Submitting print outputs for supervisor feedback	• Local / network laser printer • Bond paper supplies • Completed report folders	• Hands-on printing drill • Peer review and submission	• Printed output inspection • Competency verification

RECOMMENDED MODULE RESOURCES & EQUIPMENT (CLASS OF 30 TRAINEES)

HARDWARE & EQUIPMENT	SOFTWARE & DIGITAL MATERIALS	SUPPLIES & REFERENCE ITEMS
• 30 PC Workstations (Core i3 / 8GB RAM) • 1 Local / Network Multi-function Laser Printer • 1 Instructor Projection Unit / 55" Display • Reliable UPS / AVR Protection Units	• Licensed Spreadsheet Suite (MS Excel / LibreOffice Calc) • Standard accounting & payroll datasets • Pre-built financial exercise templates	• Letter & A4 White Bond Paper (70–80 gsm) • Spreadsheet formula quick-reference cheat-cards • Trainee Competency Progress Logbooks • Printer toner cartridges

Institutional Sign-Off & Competency Certification: Trainees must achieve satisfactory marks across all 4 learning outcomes to certify industry-standard competence in spreadsheet modeling, business calculations, chart visualization, and production printing under TESDA Basic Computer Literacy standards.